

# Farm Risk Report — NW Arkansas

Rice (50 ac), Wheat (20 ac), Soybeans (30 ac)

ENSO: Weak/Moderate La Niña • Generated in 92.7s • IBM Granite: fallback mode

| Expected Revenue | Median Loss (P50) | Worst-Case (P10) | Risk Level |
|------------------|-------------------|------------------|------------|
| \$88,376         | \$33,621 (38.0%)  | \$64,702 (73.2%) | Critical   |

## What You Should Do This Season

This season looks critical risk driven by unknown, amplified by neutral conditions. Your farm stands to lose a median of \$33,622 (38% of expected revenue) in a typical bad year, and up to \$64,702 in a worst-case scenario. The most important action right now is to buy protective put contracts on Rice and Wheat and Soybeans, costing approximately \$226 in total premium. Act before planting — options premiums rise as the season progresses and uncertainty resolves.

## Detailed Advisory

**SEASONAL CONTEXT** The current climate setup features neutral conditions. The current climate setup features a weak La Niña (ONI=-0.55). For the Delta region, this historically produces a 50% probability of above-normal temperatures and a 55% probability of below-normal precipitation during the core growing season (April–September). Both temperature and precipitation probabilities are elevated above climatological base rates (33%), indicating an increased risk of hot and dry growing conditions that could materially stress yields. Recent weather data shows a precipitation deficit of 55% below normal for the past 30 days, which compounds the seasonal dryness signal.

**PER-CROP RISK** Rice (50.0 acres): Expected revenue \$64,350. Median loss \$25,869, worst-case \$48,391. Risk level: critical. Recommendation: Buy 1 Rice put contract on CME/CBOT at a \$13.18/cwt strike. Estimated premium: \$176 total (0.3% of expected revenue). This hedge covers your downside if prices drop below \$13.18/cwt, partially offsetting a median-scenario loss of \$25,869.

Wheat (20.0 acres): Expected revenue \$5,880. Median loss \$2,364, worst-case \$3,881. Risk level: critical. Recommendation: Buy 1 Wheat put contract on CME/CBOT at a \$4.73/bushel strike. Estimated premium: \$3 total (0.1% of expected revenue). This hedge covers your downside if prices drop below \$4.73/bushel, partially offsetting a median-scenario loss of \$2,364.

Soybeans (30.0 acres): Expected revenue \$18,147. Median loss \$5,390, worst-case \$12,430. Risk level: critical. Recommendation: Buy 1 Soybeans put contract on CME/CBOT at a \$9.96/bushel strike. Estimated premium: \$46 total (0.3% of expected revenue). This hedge covers your downside if prices drop below \$9.96/bushel, partially offsetting a median-scenario loss of \$5,390.

**WHAT TO WATCH** As the season progresses, monitor soil moisture weekly. If a second consecutive dry month develops or a heat event exceeds 7 consecutive days during the critical window, consider rolling your put options to a higher strike or adding contracts. Conversely, if significant rain arrives in the first 45 days of the growing season, you may be able to sell your puts back at a profit.

## Crop-by-Crop Risk Analysis

| Rice                           | 50 acres              | Critical Risk      |               |
|--------------------------------|-----------------------|--------------------|---------------|
|                                | Revenue               | Yield %            | Loss          |
| Full season                    | \$64,350              | 100%               | —             |
| Good year (P90)                | \$52,638              | 81.8%              | (\$11,711)    |
| Median (P50)                   | \$38,481              | 59.8%              | \$25,868      |
| Bad year (P10)                 | \$15,958              | 24.8%              | \$48,391      |
| Distribution (P10 / P50 / P90) | 24.8% / 59.8% / 81.8% | $\sigma = 20.06\%$ | skew = -0.307 |

**Hedge recommendation:** Buy 1 Rice put contract on CME/CBOT at a \$13.18/cwt strike. Estimated premium: \$176 total (0.3% of expected revenue). This hedge covers your downside if prices drop below \$13.18/cwt, partially offsetting a median-scenario loss of \$25,869.

| Wheat                          | 20 acres              | Critical Risk      |               |
|--------------------------------|-----------------------|--------------------|---------------|
|                                | Revenue               | Yield %            | Loss          |
| Full season                    | \$5,879               | 100%               | —             |
| Good year (P90)                | \$4,439               | 75.5%              | (\$1,440)     |
| Median (P50)                   | \$3,516               | 59.8%              | \$2,363       |
| Bad year (P10)                 | \$1,999               | 34.0%              | \$3,880       |
| Distribution (P10 / P50 / P90) | 34.0% / 59.8% / 75.5% | $\sigma = 15.86\%$ | skew = -0.377 |

**Hedge recommendation:** Buy 1 Wheat put contract on CME/CBOT at a \$4.73/bushel strike. Estimated premium: \$3 total (0.1% of expected revenue). This hedge covers your downside if prices drop below \$4.73/bushel, partially offsetting a median-scenario loss of \$2,364.

| Soybeans                       | 30 acres              |                   | Critical Risk |
|--------------------------------|-----------------------|-------------------|---------------|
|                                | Revenue               | Yield %           | Loss          |
| Full season                    | \$18,146              | 100%              | —             |
| Good year (P90)                | \$15,696              | 86.5%             | (\$2,449)     |
| Median (P50)                   | \$12,757              | 70.3%             | \$5,389       |
| Bad year (P10)                 | \$5,716               | 31.5%             | \$12,430      |
| Distribution (P10 / P50 / P90) | 31.5% / 70.3% / 86.5% | $\sigma$ = 20.84% | skew = -0.714 |

**Hedge recommendation:** Buy 1 Soybeans put contract on CME/CBOT at a \$9.96/bushel strike. Estimated premium: \$46 total (0.3% of expected revenue). This hedge covers your downside if prices drop below \$9.96/bushel, partially offsetting a median-scenario loss of \$5,390.

Hedge Portfolio Summary

| Crop     | Strategy       | Contracts | Strike    | Premium | Cost % |
|----------|----------------|-----------|-----------|---------|--------|
| Rice     | Protective Put | 1         | \$13.1835 | \$176   | 0.3%   |
| Wheat    | Protective Put | 1         | \$4.7261  | \$3     | 0.1%   |
| Soybeans | Protective Put | 1         | \$9.9625  | \$46    | 0.3%   |
| TOTAL    |                |           |           | \$226   | 0.3%   |

Your hedge budget: \$2,651 • Recommended premium: \$226 • Budget utilization: 9%

Seasonal Climate Context

|                    |                       |                      |        |
|--------------------|-----------------------|----------------------|--------|
| ENSO Phase         | Weak/Moderate La Niña | ONI                  | -0.55  |
| 30-day anomaly     | Dry                   | Precip anomaly       | -54.9% |
| Temp anomaly       | +0.62°C               | Current dry streak   | 9 days |
| 14-day precip      | 87.7 mm               | Heat stress days     | 0      |
| Temp > normal prob | 50%                   | Precip < normal prob | 55%    |
| Drought risk       | Elevated              |                      |        |

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